

Does a Jacobian-Lens Disposition Readout Inherit the Evasion (Non)Robustness of Trained Probes? A Cross-Family Pilot

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Abstract

Trained latent-space probes are evadable by *obfuscated activations* while behavior is preserved (Bailey et al., 2024; Gupta and Jenner, 2025; McGuinness et al., 2025). The **Jacobian lens** (J-lens) is a structurally different monitor—a *fixed, unsupervised, input-averaged* Jacobian readout $J_\ell = \mathbb{E}[\partial h_{\text{final}}/\partial h_\ell]$ decoded by the unembedding—with no obfuscation study to date. We ask whether it inherits trained-probe (non)robustness. On a surface-blind factual testbed—with the dynamics findings additionally replicated across **two model families** (Qwen2.5-7B, Mistral-7B-v0.3) and a **second behavior** (refuse/comply)—we find: (i) for *detection* the J-space readout is a linear reparametrization of raw activations (no gain); (ii) the averaged Jacobian is heavy-tailed (high participation rank, low stable rank $\approx 57/3584$ even late), so a large low-energy subspace is *geometrically* available to hide in; and (iii) the **average-vs-local Jacobian gap**—the one evasion surface with no analogue for trained probes—is, against proper baselines, a **weak and noisy** single-layer attack surface: the local behavior direction is *not* concentrated where the lens is blind (energy 0.318 vs. chance 0.354) and only $\sim 2\times$ more potent there than matched random, with overlapping distributions; meanwhile behavior-flips *do* surface in the multi-layer readout (late top-1 flips 8/9–12/12), which **resists even a white-box adaptive optimization attack** (flip-and-evade succeeds 1/16). Our contribution is thus **methodological and structural**: a cautionary case where norm-ratio and full-space-random comparisons manufacture a dissociation that energy and matched-random baselines dissolve, and a first-order **pincer** motivating an average-vs-local *monitor-OOD flag* (sensitivity, not specificity; empirically unvalidated).

1 Introduction

Trained linear probes on internal activations are the standard tool for latent-space monitoring, and a growing body of work shows they can be **evaded by activation obfuscation** while black-box behavior is preserved: sparse autoencoders, probes, and latent OOD detectors all fall to obfuscated activations (Bailey et al., 2024); reinforcement learning can teach a model to evade latent monitors, with the evasion generalizing to unseen monitors of the same type (Gupta and Jenner, 2025); and a model can learn to hide from *unseen* monitors zero-shot (McGuinness et al., 2025).

The **Jacobian lens** (J-lens) is a recent alternative that reads disposition differently: it is not a *trained classifier* but a *fixed, unsupervised, input-averaged* Jacobian that linearly transports a residual into the output basis and decodes it with the model’s unembedding (Anthropic, 2026). No obfuscation study exists for it. We ask a narrow, structural question: *does a fixed averaged-Jacobian disposition readout inherit the evasion (non)robustness known for trained probes—and if not, why?*

We do **not** re-discover that holistic/late monitors resist obfuscation because behavior must surface—that is the stated core insight of Gupta and Jenner (2025). Nor do we claim a new attack on latent monitors (Bailey et al., 2024; McGuinness et al., 2025). Our contributions occupy the slices those works structurally cannot:

1. **The object.** All prior work attacks *trained* probes/SAEs/classifiers. The J-lens is fixed, unsupervised, and averaged-Jacobian; whether it is more or less evadable was untested. We give the first obfuscation-style probe of it.
2. **A spectral observation** consistent with (not a derivation of) the must-surface bound: the layerwise effective-rank profile is a *candidate*

mechanism, while our own leaks show rank alone does not force a top- k flip.

3. **A methodological caution + a design principle**, where the paper’s weight lands. We target the average-vs-local gap, find it *weak and noisy* under proper baselines, and distill a first-order **pincer** motivating a monitor-OOD flag.

2 Related work

Obfuscation attacks. Bailey et al. (2024) is the foundational attack: SAEs, representation probes, and latent OOD detectors are all bypassable, dropping monitor recall 100% $\rightarrow$ 0% at a 90% jailbreak rate, with a capability tax on complex tasks. Gupta and Jenner (2025) finetune 7B–14B models via RL to evade latent monitors while preserving behavior; token-level monitors are easily evaded, but *holistic* monitors (max-pool, attention) remain robust, and evasion generalizes across monitors of a type. McGuinness et al. (2025) show zero-shot evasion of *unseen* monitors, with obfuscation localized near the probe layer (early-to-mid ramp, late decay), and propose detecting low-rank activation shifts as a defense.

Positioning. Three prior results directly abut ours and must be distinguished, not re-derived: (a) robustness of holistic/late monitors is the empirical claim of Gupta and Jenner (2025)—we offer a candidate *spectral observation* consistent with it, not the claim; (b) detecting low-rank shifts as a defense is proposed by McGuinness et al. (2025)—our flag keys on *average-vs-local Jacobian divergence*, a quantity specific to averaged-Jacobian readouts; (c) early-vs-late localization is observed by McGuinness et al. (2025)—we relate it to the averaged Jacobian’s effective-rank profile.

Adjacent. Nordby et al. (2026) shows probe accuracy scales with size and benefits from multi-layer ensembling (clean accuracy only; disclaims adversarial robustness). Blandfort and Graham (2025) red-teams probes with prompted LLMs; Fonseca Rivera and Africa (2025) is on the defense side. We adopt evaluation cautions from Kretschmar et al. (2025); Smith et al. (2025), and our refuse/comply setup follows the refusal-direction methodology of Arditi et al. (2024).

Gradient- and activation-based detectors. A parallel line builds runtime jailbreak detectors from gradients, activations, or logits: refusal-loss

gradient norms (Hu et al., 2024), hidden-state monitors (Jiang et al., 2025; Mia et al., 2026), and token-level logit signals (Zhang et al., 2026). Our $D(x)$ (§5) is in this family but keys specifically on *average-vs-local Jacobian disagreement*; the token-level Jacobian-attribution toolkit of Liu et al. (2026) (causal attributions via information geometry) is a natural source of the richer $D(x)$ metrics our geometry-aware variant (§4.4) begins to exploit. These detectors target a different problem (input-jailbreak detection) than our threat model (activation-obfuscation robustness of a fixed readout); a head-to-head runtime comparison is future work. A related line builds *multi-layer* representation-trajectory fingerprints, robust across layers and positions (Kadali and Papalexakis, 2026; Ball et al., 2026); our multi-layer $D(x)$ (§4.4) tests this idea for the average-vs-local divergence but—consistent with our small- n caveats—does not yield a reliable detector at our scale.

3 Threat model and setup

J-lens. $\text{lens}_\ell(h) = \text{unembed}(J_\ell h)$, with $J_\ell = \mathbb{E}[\partial h_{\text{final}}/\partial h_\ell]$ averaged over a corpus of prompts and positions. A monitor reads the **discrete top- k** of this readout.

Threat model. A white-box, activation-space attacker perturbs the residual stream to change behavior while keeping the monitor’s readout benign; for a top- k monitor, “benign” means the top- k does not flip. This isolates the *geometry* of the readout operator.

Model & lens. Qwen2.5-7B-Instruct (bf16, 28 layers); lens fit on 85 generic prompts ($d_{\text{model}}=3584$, source layers 0–26). §4.5 adds Mistral-7B-Instruct-v0.3.

Surface-blind testbed. Factual statements “{subject} is {object}”; false items via a **derangement** of objects within a category, so every token appears equally often as true and false. A surface classifier (length; TF-IDF 1–2gram) under **leave-category-out** scores AUROC 0.500. 68 items, 5 categories; all probes evaluated leave-category-out.

Estimation and definitions (reproducibility). *Averaged Jacobian:* we use the reference jlens estimator— J_ℓ is the empirical mean of $\partial h_{\text{final}}/\partial h_\ell$ over generic pretraining-like prompts, source positions, and current-and-future target po-

metric	early ($\ell \leq 6$)	late ($\ell \geq 20$)
participation rank $(\sum \sigma)^2 / \sum \sigma^2$	~ 826	~ 2919
stable rank $\ J\ _F^2 / \ J\ _2^2$	~ 8	~ 57

Table 1: Threshold-free effective rank of J_ℓ (Qwen, $d=3584$). The early $\rightarrow$ late rise survives both metrics, but the spectrum is heavy-tailed: stable rank ~ 57 of 3584 even late.

sitions (the model’s own backward pass), stored as a $d \times d$ matrix per source layer. Our lens averages 85 prompts (a checkpoint of a 100-prompt fit; the reference recommends ~ 100 –1000); corpus-size sensitivity is untested (§6). *Low-energy* (“blind”) *subspace*: we SVD $J_\ell = U\Sigma V^\top$ and split V at the smallest r_{99} capturing 99% of spectral energy $\sum \sigma^2$; the *visible* subspace is $\text{span}(V_{:r_{99}})$ and the *blind* subspace its orthogonal complement ($d - r_{99}$ directions). “Energy in the blind subspace” of v is $\|P_{\text{blind}}v\|^2 / \|v\|^2$; the chance baseline is $k/d = (d - r_{99})/d$. *Surfacing*: for a steered input we read $\text{unembed}(J_{\ell'}h_{\ell'})$ at a late layer $\ell'=26$ (unless noted) and call a top- k “surfacing event” any change in the top- k token set vs. baseline (top-1 = argmax flip; top-3/5 = set symmetric difference); a “top-1 miss” is an unchanged argmax with a moved top- k set.

4 Results

4.1 Detection is a linear reparametrization

Internal true/false AUROC is **1.000** ($\ell \geq 15$) vs. surface 0.500; a 1–3-dimensional direction suffices. Since $J_\ell h$ is a *linear* map of h , a linear probe on it cannot beat one on h . At sub-ceiling layers (where the comparison is informative) raw and J-space probes have *overlapping* bootstrap CIs with raw $\geq$ J-space—e.g., $\ell=8$: raw 0.876 [0.78, 0.96] vs. J-space 0.780 [0.66, 0.89] (leave-category-out)—confirming the readout is a *lossy linear reparameterization* with no detection gain. We pivot to evasion-robustness.

4.2 Spectral texture: geometry offers hiding room

Table 1 shows the averaged Jacobian is heavy-tailed: a large low-energy subspace—invisible to the readout by construction ($J_\ell \delta \approx 0$)—is *geometrically* available. Geometry alone would suggest the lens is evadable.

4.3 Dynamics: the near-theorem

Empirical near-theorem. A behaviorally-effective perturbation must influence the output; the output is a function of the late residual; the late readout is high-participation-rank over that residual; hence behavior change $\Rightarrow$ the late readout changes $\Rightarrow$ visible—*modulo the average-vs-local gap* (§4.4), which is the “near.”

Test. We build the behavior direction (true $\rightarrow$ false mean difference) at $\ell=15$, split it into visible (row-space of J_{15}) and hidden (low-energy) parts, steer with each, and use a random control. The averaged mean-difference hidden direction is as blunt as random (flips only at $\alpha \approx 2\|h\|$); but this is a property of that averaged direction, not the subspace (§4.4). Among behavior-flip cases, $P(\text{late top-1 flips} \mid \text{behavior flips}) = 8/9$; the single leak (behavior changed, late top-1 did not) sits at a *wide* margin (3.12). So the multi-layer readout *appears* robust (small n ; §6) with a top- k -slack asterisk—not a clean 12/12, and the leak is unstable across runs. At larger $n=34$ (all truth prompts) the estimate tightens: late surfacing is top-1 32/34 = **0.94** (bootstrap 95% CI [0.85, 1.00]) and **top-3/top-5 34/34 = 1.00**, showing the top-1 misses are top- k -threshold artifacts rather than genuine multi-layer evasions.

4.4 The average-vs-local gap, deflated

The near-theorem uses the *averaged* J_ℓ ; behavior is governed by the *local* Jacobian $\partial F / \partial h$ at the attack input. A clean spoof would live in a direction low-energy in the averaged map (lens-blind) but high-energy in the local map (behaviorally live)—a question structurally unique to averaged-Jacobian readouts. Per input we take the local behavior gradient $g(x) = \partial(\ell_{\text{false}} - \ell_{\text{true}}) / \partial h_{15}$, its averaged-hidden component, and its steering potency. Against the baselines a reviewer would demand:

- **No energy concentration.** g has **0.318** of its *energy* in the averaged-hidden subspace, *below* the random baseline $k/d = 1269/3584 = 0.354$. (The eye-catching “57%” is a *norm* ratio; in energy, against chance, there is no concentration.)
- **Modest, noisy potency.** Local-hidden flips at median $\alpha=0.375\|h\|$ vs. a matched **random-hidden** baseline of $0.75\|h\|$ — $\sim 2\times$, not the $8\times$ obtained by comparing to a full-space random direction; per-input distributions over-

lap.

- **Surfacing strong, leaks unstable.** Late top-1 flips 12/12 in this run; 8/9 and 11/12 elsewhere.

So the J-lens does **not** cleanly inherit a single-layer evasion; the finding is *robustness with nuance*. We report the retracted strong claim to document how a norm-ratio + full-space-random comparison manufactures an effect that energy + matched-random baselines remove.

4.5 Replication across families and behaviors

Second family (Mistral-7B-v0.3). On a different-family 7B (32 layers, $d=4096$, own lens; INJ 17, LATE 30) all three replicate: spectrum heavy-tailed (participation $1040 \rightarrow 3998$; stable rank $\sim 14 \rightarrow 154$, i.e. 3.8% of d even late); behavior-flips surface late **12/12**; local-gradient blind-subspace energy **0.110** < **chance 0.150** (no concentration), noisy potency lean $\sim 4\times$ (median α 0.5 vs. random-hidden 2.0).

Second behavior (refuse/comply, Qwen). Using a standard refusal-direction setup (Arditi et al., 2024)—matched-format instructions, refusal-triggering vs. benign, *non-operational* (only requests are written; internal state / next-token read; no harmful output generated)—which is *not* surface-blind, so only the dynamics replicate: steering the comply direction flips the next token **12/12** and surfaces late **11/12**; blind-subspace energy **0.349** $\approx$ **chance 0.354**, with a *cleaner* $\geq 4\times$ potency lean (all 12 at the α floor 0.25 vs. random-hidden 1.0).

Larger curated adversarial set. To rule out small hand-written testbeds, we re-ran the dynamics on **35 prompt-injection / jailbreak prompts from Giskard’s bundled adversarial dataset** (vs. 35 benign instructions; dynamics only). All replicate at the larger n : comply-steer flips behavior **34/35** and surfaces late **34/35** (97%); blind-subspace energy **0.357** $\approx$ **chance 0.354**; potency lean median α 0.25 vs. random-hidden 1.5 ($\sim 6\times$). Consolidating, the potency lean grows with behavior ($\sim 2\times$ truth, $\sim 4\times$ refuse/comply, $\sim 6\times$ injection), but the robust invariant is *energy at chance* plus high multi-layer surfacing, across two models, three behaviors, and n up to 35.

Robustness of layer choice. For L15 injection, surfacing is not specific to L26: behavior-flips surface in the late top-1 at 12/12 (L18–L24) and

attack ($n=16$)	flip	readout intact	success
naive (null-space)	14/16	1/16	0/16
behavior-only	16/16	0/16	0/16
adaptive	11/16	3/16	1/16

Table 2: A white-box *optimization* attack (δ at h_{15} , $\|\delta\| \leq 2\|h\|$) cannot flip behavior while keeping the late (L26) readout intact (success = flip $\wedge$ intact). The evasion penalty trades off against flipping.

11/12 (L26). The pattern—heavy-tailed operator, robust multi-layer surfacing, weak-and-noisy single-layer gap—holds across two families and two behaviors; only the potency-lean size varies ($\sim 2\times$ noisy for truth, $\sim 4\times$ for Mistral/refuse-comply).

4.6 An adaptive attack does not evade the readout

Our attacks so far are non-adaptive (null-space steering, blind to the lens). We now let a white-box attacker *optimize* against the monitor (obfuscated-activations style, Bailey et al., 2024): per input we run 60 Adam steps on a perturbation δ at h_{15} (norm-capped at $2\|h\|$) that maximizes the behavior flip *and* penalizes any change in the late (L26) readout logits. Even so, flip-*and*-evade succeeds in only **1/16** cases (Table 2): the naive steer flips 14/16 but keeps the readout intact only 1/16; optimizing behavior alone flips 16/16 but never keeps the readout intact; and adding the evasion penalty mostly *reduces* the flip rate ($16 \rightarrow 11$) rather than buying stealth. This tension is the near-theorem’s teeth—keeping the late readout fixed forces the attacker to stop changing behavior. So the multi-layer readout resists not only naive but *adaptive* (optimization-based) attacks, with a small (1/16) residual consistent with the top- k slack. Full RL-Obfuscation-style *learned* attacks that generalize across inputs (Gupta and Jenner, 2025; McGuinness et al., 2025) remain future work.

5 Application: a monitor-OOD flag

The average-vs-local divergence $D(x)$ is monitorable at runtime: if $D(x) > \tau$, flag not “harmful” but “*this monitor’s verdict is untrustworthy here*,” triggering a fallback (escalate/abstain/review)—complementary to holistic monitors (Gupta and Jenner, 2025).

A first-order pincer. A single-layer evasion needs a direction that is (a) locally potent and (b)

averaged-lens-hidden; but (b) is low energy in the averaged Jacobian and (a) is high energy in the local one, and their disagreement is $D(x)$. Hence *potent+hidden* $\Rightarrow$ high $D(x)$ $\Rightarrow$ the flag fires, while *potent+visible* $\Rightarrow$ the single-layer monitor catches it directly. To first order there is no third option.

Sensitivity, not specificity—measured. The pincer is one-sided: it gives *sensitivity* (attack $\Rightarrow$ high D), not *specificity*. We tested the cheapest $D(x)$ (energy fraction of the local gradient in the blind subspace) across $n=34$ inputs: it is essentially constant (at chance 0.354) and predicts attackability at AUROC **0.548** ($\approx$ chance). We also tried richer proxies. A *geometry-aware* variant—the local map’s finite-difference response (JVP) on averaged-*blind* vs. full input directions—is **unstable across runs** (AUROC 0.41–0.64 in two draws; bootstrap 95% CI [0.13, 0.61] at $n=34$, straddling chance), and **aggregating it across four layers does not help** (multi-layer AUROC 0.36). So all three proxies—cheap, geometry-aware, and multi-layer—fail to reliably separate attackable inputs at this scale. The pincer therefore stands as a *design principle with a first-order guarantee*, but we do **not** have a working detector; realizing one (e.g., with richer Jacobian attributions (Liu et al., 2026) or multi-layer refusal-trajectory fingerprints (Kadali and Papalexakis, 2026; Ball et al., 2026)) and validating it against a benign contrast is the central open problem. Cost would be a few JVPs per monitored layer, reducible via low-rank projection.

6 Conclusion

In a preliminary pilot that replicates across two model families (Qwen2.5-7B, Mistral-7B-v0.3) and two behaviors (factual truth; refuse/comply), a fixed unsupervised averaged-Jacobian disposition readout does **not** cleanly inherit trained-probe evadability: the probe-untouchable average-vs-local gap is real but weak under proper baselines, and behavior-flips reliably surface in the multi-layer readout—which resists even a white-box adaptive optimization attack (flip-and-evade 1/16). The durable contributions are a transferable methodological caution (energy + matched-random baselines dissolve an apparent dissociation) and a first-order pincer motivating an average-vs-local monitor-OOD flag—a design principle awaiting benign-separation validation and a harder testbed.

Limitations

The single-layer evasion is weak and noisy (§4.4): energy at/below chance, $\sim 2\text{--}4\times$ potency lean over matched random with overlapping per-input distributions at $n=12$. Multi-layer surfacing is small- n : 11/12 gives a wide Wilson 95% CI $\approx$ [0.65, 0.98], 8/9 $\approx$ [0.52, 0.99]; we therefore say the readout *appears* robust, not *is*. The pincer is stated for the *linearized* (Jacobian) regime, though our adaptive optimization attack (§4.6) already probes beyond first order and the readout held (1/16); a fully *learned*, weight-space RL-Obfuscation attack that generalizes across inputs is untested and could in principle differ. The flag $D(x)$ does not reliably separate: the cheap proxy is at chance (0.548), a geometry-aware variant is unstable across runs (0.41–0.64, wide CIs), and a multi-layer aggregate does not help (0.36)—so we have a design principle, not a working detector (§4.4). The core pattern replicates across two families and three behaviors (truth, refuse/comply, and a 35-prompt Giskard injection set), and the readout-layer choice is shown non-critical; what remains is that only the truth condition is surface-blind (detection replicates there only), the truth testbed is too easy (internal AUROC 1.000), lenses are below the tool’s recommended prompt count, and the potency lean uses coarse α steps (a lower bound for refuse/comply and injection). We treat this as a preliminary, multiply-replicated pilot rather than an archival claim.

Ethical Considerations

Our refuse/comply analysis follows the established refusal-direction methodology (Arditi et al., 2024). It is *non-operational*: we write only refusal-triggering *requests* (no harmful instructions), and we read internal activations and the next-token distribution rather than generating harmful content. The work is defensive in aim—characterizing when an internal safety monitor’s verdict should not be trusted—and uses only open-weights models and public methods.

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